

tf.compat.v1.metrics.accuracy Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/metrics/accuracy

Calculates how often predictions matches labels.

Function Signatures

```
tf.compat.v1.metrics.accuracy( labels, predictions,  
weights=None, metrics_collections=None,  
updates_collections=None, name=None )
```

Code Examples

```
tf.compat.v1.metrics.accuracy(  
labels,  
predictions,  
weights=None,  
metrics_collections=None,  
updates_collections=None,  
name=None  
)
```

```
tf.compat.v1.metrics.accuracy(  
    labels,  
    predictions,  
    weights=None,  
    metrics_collections=None,  
    updates_collections=None,  
    name=None  
)
```

```
accuracy, update_op = tf.compat.v1.metrics.accuracy(  
    labels=labels,  
    predictions=predictions,  
    weights=weights,  
    metrics_collections=metrics_collections,  
    update_collections=update_collections,  
    name=name)
```

```
accuracy, update_op = tf.compat.v1.metrics.accuracy(  
    labels=labels,  
    predictions=predictions,  
    weights=weights,  
    metrics_collections=metrics_collections,  
    update_collections=update_collections,  
    name=name)
```

```
m = tf.keras.metrics.Accuracy(  
    name=name,  
    dtype=None)  
m.update_state(  
    y_true=labels,  
    y_pred=predictions,  
    sample_weight=weights)  
accuracy = m.result()
```

```
m = tf.keras.metrics.Accuracy(  
    name=name,  
    dtype=None)  
m.update_state(  
    y_true=labels,  
    y_pred=predictions,  
    sample_weight=weights)  
accuracy = m.result()
```

```
g = tf.Graph()  
with g.as_default():  
    logits = [1, 2, 3]  
    labels = [0, 2, 3]  
    acc, acc_op = tf.compat.v1.metrics.accuracy(logits, labels)  
    global_init = tf.compat.v1.global_variables_initializer()  
    local_init = tf.compat.v1.local_variables_initializer()  
    sess = tf.compat.v1.Session(graph=g)  
    sess.run([global_init, local_init])  
    print(sess.run([acc, acc_op]))  
[0.0, 0.66667]
```

```
g = tf.Graph()
```

Documentation

Calculates how often predictions matches labels.

Migrate to TF2

`tf.compat.v1.metrics.accuracy` is not compatible with eager execution or `tf.function`.

Please use `tf.keras.metrics.Accuracy` instead for TF2 migration. After instantiating a `tf.keras.metrics.Accuracy` object, you can first call the `update_state()` method to record the prediction/labels, and then call the

result() method to get the accuracy eagerly. You can also attach it to a Keras model when calling the compile method. Please refer to this guide for more details.

Structural Mapping to Native TF2

Before:

After:

How to Map Arguments

Before & After Usage Example

Before:

After:

Description

Used in the notebooks

- Migrate metrics and optimizers
- Exploring the TF-Hub CORD-19 Swivel Embeddings
- Graph-based Neural Structured Learning in TFX

The accuracy function creates two local variables, total and count that are used to compute the frequency with which predictions matches labels. This frequency is ultimately returned as accuracy: an idempotent operation that simply divides total by count.

For estimation of the metric over a stream of data, the function creates an `update_op` operation that updates these variables and returns the accuracy. Internally, an `is_correct` operation computes a Tensor with elements 1.0 where the corresponding elements of predictions and labels match and 0.0 otherwise. Then `update_op` increments total with the reduced sum of the product of weights and `is_correct`, and it increments count with the reduced sum of weights.

If weights is None, weights default to 1. Use weights of 0 to mask values.

Returns

`## Raises`